SESSION 2: EARNINGS BASED METHODS

Learning Objectives

At the end of this session, you should be able to:

a) determine the value of a share of a business using the price earning based method,
b) determine the value of a share of a business using the accounting rate of return [ARR] method, and
c) calculate the value of a share of a business using the discounted cash flow [DCF] method.

Introduction

Earnings based methods are derived from income statement and market values. The value of a share or business is based on its ability to generate profits. The fundamental idea is that the value of a business depends on the future earnings it can generate for investors. The methods may refer to past, present or future earnings, and after relevant adjustments as may be required, to form the basis of estimation of the value of the business. This session focuses on three main earnings-based methods, namely Earnings Yield/Price Earnings Method, Accounting Rate of Return and Discounted Cashflows.

The Earnings Yield /Price Earnings Method

Price Earnings Ratio (PER) method is similar to the Earnings Yield (EY) method. One is the inverse of the other. PER = MP/EPS and EY = EPS/MP or 1/PER.

This method is appropriate in valuing a controlling interest in a company where the owner can decide on dividend retention policy. The value of a share is based on the entity's earnings per share and an appropriate P/E ratio, which is market or industry based. The P/E ratio relates earnings per share to the value of share. The earnings may be historical, current or estimated future earnings. The preferred EPS however is one derived from current or future earnings.

Since P/E = Market price per Share (MPS)
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Earnings per share (EPS)

Market Price per share = P/E ratio x EPS

Since EY = EPS/MPS, **MPS = EPS/EY**

When a P/E Ratio is given and the question insists on using Earnings Yield to compute share price, compute the EY as the inverse of the P/E ratio given. That is 1 divided by the P/E Ratio. Example if PER = 6 times, EY = 1/6.
P/E ratio of 15 = an EY of 0.0667
P/E ratio of 7 = an EY of 0.1429
P/E ratio of 10 = an EY of 0.10

Price Earnings Ratio

The value of a share of a company is arrived at by deciding on a suitable P/E ratio and multiplying this by the EPS for the shares which are being valued. For a given EPS figure, a higher P/E ratio will result in a higher share valuation. A higher P/E ratio may indicate the following:

- Expectation that the EPS will grow rapidly in the succeeding years so that a high price is being paid for future profit prospects.
- There is a high security of earnings i.e. a well-established low risk company is valued on a higher P/E ratio than a similar company whose earnings are subject to greater uncertainty.

This means that since unquoted companies are unlisted, their risk associated factors are high making them at risk of high uncertainty to achieve projected EPS. Hence, they tend to have relatively low P/E ratios compared to quoted or listed companies.

The following points should guide the use of the method:

- When both historical and current EPS data are provided, it is advisable to communicate a range of share values.
- The average of all the EPS given (including the current year's) should be used to compute the share value using the P/E ratio determined.
- Using the same P/E ratio, compute the share value based on only the current EPS.
- The possible share price of the company should be communicated as a range of the two values ascertained.

When a company is thinking of acquiring an unquoted company in a take-over, the final offer price will be agreed by negotiation.

Illustration 1

ABC Ltd wishes to make a take-over for the shares of an unquoted company DEF company. The earnings of DEF for the past five years are as follows:

	GH¢m
2021	200
2022	288
2023	272
2024	284
2025	300

The average P/E ratio of quoted companies in the industry in which DEF operates is 10. Quoted companies which are similar in many respects to DEF are

QRS which has a P/E ratio of 15 but is a company with very good growth prospects

JKL which has a poor profit record for several years and has a P/E ratio of 7

What would be the suitable range of valuations for the shares of DEF?

Solution 1

Price per share = EPS X P/E ratio

Earnings:

Average earnings over the last five years has been ¢268.8 million and the most recent year is ¢300m. There appears to be a growth prospect.

P/E Ratio:
P/E ratio of 15 (QRS) would be much too high for DEF since the growth prospect is not as certain and is also unquoted. On the other hand, DEF's expectation of earnings is probably better than JKL (7). A suitable P/E ratio might be based on the industry's (10) but since DEF is not listed, and therefore riskier, a lower P/E ratio may be more appropriate; perhaps 80% of 10 = 8
The valuation of DEF's shares might therefore range between

- 8 x ¢300m = ¢2,400 million, and
- 8 x ¢268.8 = ¢2,150.4 million

When a company is thinking about taking over another, it should look at the target company's forecast earnings and not historical earnings. However, forecast of earnings growth should only be used if:

- there are good reasons to believe that earning growth will be achieved;
- A reasonable estimate of growth rate can be made;
- Any forecast supplied by the target company's Board of Directors are made in good faith

Problems associated with using the Earnings yield/P/E ratio basis include:

- The calculation of earnings figure is arbitrary;
- Selecting a suitable P/E ratio is difficult;
- Any discounting of the selected P/E ratio of a quoted company is to an extent, arbitrary.

Illustration 2
Pass Ltd is performing extremely well in the manufacturing industry and is now considering listing on the stock market. The company is currently capitalized with 5 million ordinary shares. A summary of the current financial statement extract is provided below.

	GH¢'million
Profit before tax	24
Taxation	(8.4)
Profits after tax	15.6

Pass Ltd.'s management has obtained some financial information on companies listed in the same industry. Walk Ltd is performing at par as Pass Ltd in the manufacturing industry.

	Walk Ltd	Sim Ltd
Market Price of Share	GH¢70	GH¢8.4
Earnings per share	GH¢4.60	GH¢0.55

Assume it is 15% percent riskier investing in an unlisted company.
Required: Calculate the possible value a share of Pass should be sold for and the total market capitalization if all shares are issued.

Solution 2

$Value\ of\ Share = PE\ Ratio \times EPS$

P/E Ratio

The P/E ratio of Pass should be based on the P/E ratio of Walk, because of the phrase "Walk Ltd is performing at par as Pass Ltd in the manufacturing industry".

Therefore, the P/E ratio of only Walk ltd will be computed and adjusted to reflect that on Pass Ltd

Walk Ltd's P/E Ratio will be $P/E = (Market\ Price\ per\ Share\ (MPS))/(Earnings\ Per\ Share\ (EPS)) = GHc70/GHc4.60 = 15.22$

Because Pass Ltd is unlisted, the P/E ratio of Walk must be adjusted to reflect the lack of marketability and risk of uncertainty. The question provided a risk factor of 15%, hence we cannot assume any adjusting percentage than to use 15%.

The P/E Ratio of Pass Ltd will therefore be adjusted to 12.94 (85% of 15.22)

Earnings Per Share

The earnings per share will be calculated as total earnings (profit after tax and preference dividends) divided by number of shareholdings.

$EPS = (Total\ Earnings\ (Profit\ after\ tax))/(Number\ of\ Shareholdings) = GHc15.6 million/(5\ million\ shares) = GHc3.12\ per\ share$

Value of Share

$Value\ of\ Share = PE\ Ratio \times EPS$

$= 12.94 \times GHc3.12\ per\ share$

$= GHc\ 40.37\ per\ share$

$Therefore\ the\ value\ of\ a\ share\ in\ Pass\ Ltd\ is\ GHc\ 40.37\ per\ share$

Total Market Capitalization

$Market\ Capitalization = Share\ Price \times Number\ of\ Shareholdings$

$= GHc\ 40.37\ per\ share \times 5\ million\ Shares$

$= GHc\ 201.85\ million$

The total market capitalization of all shares issued in Pass Ltd will be worth GHc 201.85 million

Accounting Rate of Return (ARR) Method

This method considers the accounting rate of return which will be required from the company whose shares are to be valued (It is different from the P/E method which is concerned with the market rate of return required)

$$\text{Value of a business} = \frac{\text{Estimated Future Profit}}{\text{Required Return on capital employed}}$$

For a takeover bid valuation, it is necessary to adjust the profit figure to allow for expected changes after the take-over. These might include:

1. New levels of directors' remuneration

2. New levels of interest charges (perhaps because the predator company will be able to replace existing loans with new loans at a lower rate of interest or because the previous owners had lent the company money at non-commercial rates).
3. A charge for notional rent where it is intended to sell existing properties or where the rate of return used is based on the results of similar companies that do not own other own properties

Illustration 3

A Ltd is considering acquiring B Ltd. At present B Ltd is earning an average of GH¢960m after tax. The directors of A Ltd feel that after reorganization, this figure could be increased to GH¢1,200m. All the companies in the A Ltd's group are expected to yield a post-tax return of 40% on capital employed. What should be the valuation of B Ltd?

Solution 3

$$\text{Valuation} = \frac{\text{GH¢1,200 m}}{0.4} = \text{GH¢3,000m}$$

This is the maximum that A Ltd should be prepared to pay. The initial offer would probably be lower.

The ARR valuation might be used in a takeover when the acquiring company is trying to assess the maximum amount it can afford to pay. This is because it is a measure of management efficiency and the rate used can be selected to reflect, among other things the returns which the acquiring company thinks should be obtainable after any post-acquisition reorganization has been completed. A valuation on this basis should then be compared with the stock market price (for quoted companies) or a price arrived at using the P/E ratio of similar quoted companies

Problems associated with this method include the following:

- It is difficult to calculate an accurate figure for post-acquisition profits, having taking into account the changing cost profile following the take-over
- The method does not take account of future growth in profits.

Discounted Cash Flows

This method of share valuation may be appropriate when one company intends to buy the assets of another company and to make further investments in order to improve profits in future.

Illustration 4

Wide Ltd wishes to make a bid for Narrow Ltd. Narrow Ltd makes after –tax profits of GH¢12,000m a year. Wide Ltd believes that if further money is invested in Narrow Ltd the after tax cash flows (ignoring the purchase consideration) could be as follows:

		GH¢m
Year	0	(30,000)
	1	6,000
	2	18,000
	3	30,000
	4	45,000

5	45,000

The after-tax cost of capital of Wide Ltd is 30% and the company expects all its investments to pay back in discounted terms within 5 years.
What is the maximum price that the company would be willing to pay for the shares of Narrow Ltd?

Solution 4
The maximum price is one which would make the return from the total investment exactly 30% over five years so that the NPV would be zero.

Yr	Cash Flows GH¢m	Disc, Factor	Present Value GH¢m
0	(30,000)	1	(30,000)
1	15,000	0.769	11,535
2	18,000	0.592	10,656
3	30,000	0.455	13,650
4	45,000	0.350	15,750
4	45,000	0.269	12,105

Maximum purchase price			33,696
			======

This method has some practical difficulties especially, with predicting future cash flows and choosing the appropriate discount rates